6 **(a)** State what is meant by an electric current.

....

.... [1]

(b) A cell has electromotive force (e.m.f.) 1.50 V and internal resistance 0.75 Ω.

On Figure 6.1, draw the variation of the terminal potential difference (p.d.) with current in the cell for current between 0 A and 1.6 A.

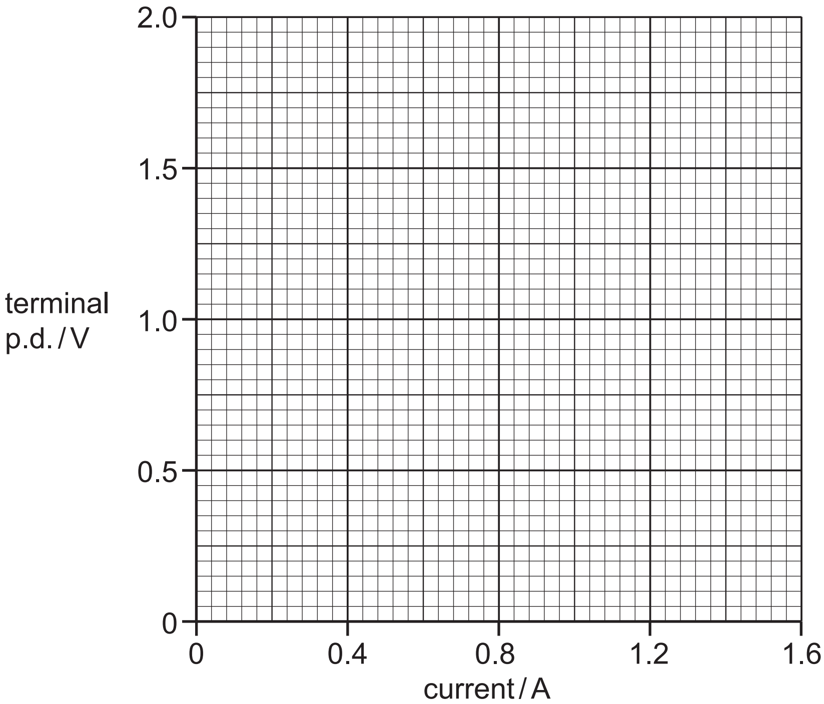

Figure 6.1

[2]

(c) A metal wire X has length 6.8 m and cross-sectional area $2.0 \times 10^{-7}\,\text{m}^2$. The resistance of the wire is 0.82 Ω.

The wire is connected to a circuit containing the cell in **6(b)** and a fixed resistor of resistance *R*, as shown in Figure 6.2.

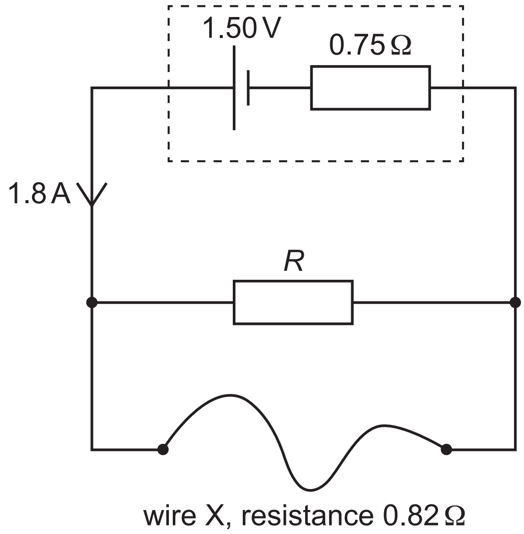

Figure 6.2

The current in the cell is 1.8 A.

(i) Calculate the resistivity ρ of the metal from which wire X is made.

ρ = ... Ω m [2]

(ii) Calculate *R*.

R = ... Ω [3]

(iii) Calculate the power dissipated in the internal resistance of the cell.

power = ... W [2]

[Total: 10]